Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one detail that helps explain a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your answer. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

An object moves **0.3 m** toward a fixed plane mirror. **Find image movement** and the change in object-image separation.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

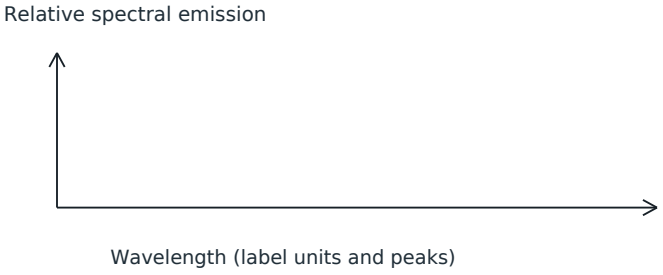
02 Worked example: explain the prediction

Reference: $\lambda_{max}\,T = \mathbf{2.9\times10^{-3}\ m\cdot K}$.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\,T \approx \mathbf{2.9\times10^{-3}\ m\cdot K}$ gives $\lambda_{peak} \approx \mathbf{9.7}$ micrometers at **300 K**.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

Two ideal thermal emitters are at **3000** and **6000 K**. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

04 Apply the physics

1. **Compare ideal thermal emitters** at **5000 K** and **10,000 K**.
2. Use $\lambda_{peak}\, T = \mathbf{2.9 \times 10^{-3}\ m{\cdot}K}$ to **calculate peak wavelengths** in nm.
3. **Sketch both broad spectra**.
4. **Explain why their peaks alone** cannot predict an exact observed meteor color.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An object has an infrared peak but looks red-hot. **Explain how both can be** true.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Across a stationary boundary, frequency remains **4 Hz** while speed changes from **24** to **32 m/s**. **Find both wavelengths**.
Label a wave sketch to **support the calculation or comparison**.

Word help: boundary = the place where a medium or constraint changes

Sentence starter: I found ... [unit]. This value means ... in this situation.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

An ideal blackbody has λ_{max}=**725 nm**. **Estimate temperature**.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find peak wavelengths** at **3000** and **6000 K**.
2. **State why wavelength peak alone** does not specify one exact perceived color.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3